
Registered but Not Anticipated: The Edit-Response Geometry of a Frozen Genomic Foundation Model

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<https://github.com/AbdelStark/GenoLeWM>

Abstract

Latent world models promise to predict the consequence of an action in representation space, skipping the encoder forward pass that would otherwise be needed to observe it. We ask whether that promise holds for the most economical genomic action—a single-base edit—on frozen DNA foundation-model features, and we find that it does not, for a reason that is information-theoretic rather than architectural. We measure the *edit-response geometry*: the displacement $\Delta = s(w_{\text{alt}}) - s(w_{\text{ref}})$ a point edit induces in a pooled state, over 12,993 GRCh38-grounded ClinVar and BRCA2 saturation-editing variants, across a pooling grid, on three encoders that share neither objective nor tokenizer. Three findings follow. First, the *predict-no-change trap* is universal: a single edit leaves $\cos(s(w_{\text{ref}}), s(w_{\text{alt}})) \geq 0.993$ on every encoder and every pooling width, reaching 1.00000 under global pooling, so an identity map wins any cosine-supervised objective—yet on Carbon-500M the pathogenicity AUROC of $\|\Delta\|$ stays flat at 0.92–0.93 while the relative displacement varies $9.4\times$. Dilution is not information loss; the trap is the metric, not the geometry. Second, whether that geometry is *informative* is an encoder-specific property, not a property of genomic FMs: the same measurement gives AUROC 0.93 on Carbon-500M, 0.76 on Nucleotide Transformer v2, and 0.55—near chance—on HyenaDNA. On Carbon-500M the response is interventional, exceeding a reference-only probe that never sees the edit by 0.13–0.31 and surviving conditioning on the full trinucleotide context (-0.006); on Nucleotide Transformer v2 at the same span it does not, the unedited window predicting better than the edit response. Third, and independent of the first two, the pathogenicity-bearing component of the response is precisely the component that cannot be anticipated: predicting Δ from the pooled reference state and the action reaches AUROC 0.652 against a 0.926 ceiling, barely above the 0.631 obtainable from the substitution class with no model at all, and the pooled state is not merely uninformative but harmful (ridge 0.585, below a lookup that ignores it). The failure deepens as the state is pooled more, as an information-bottleneck account predicts. On Carbon-500M the geometry is not a restatement of the model’s own likelihood: it beats Δ log-likelihood on identical variants (0.9255 vs. 0.8945), correlates with it only weakly ($\rho = -0.24$), retains 0.853 once it is residualised out, and ensembles to 0.948. We report these AUROCs as cohort-dependent— Δ log-likelihood reaches 0.89 on our ClinVar slice against 0.6–0.8 in the literature, so the slice is easy—and we make no claim to a competitive variant-effect predictor; transfer to quantitative function is weak ($\rho = -0.27$). These results supersede our withdrawn v0.2.1-r1 release, whose implementation defects were real but incidental: the bottleneck we identify would have defeated a correct implementation too. For an action-conditioned latent world model over pooled genomic features, there is no shortcut through the latent—the encoder must be run on the edited sequence.

1 Introduction

DNA foundation models [9, 14, 20, 21, 25] answer a representational question: *what is the embedding, or the likelihood, of this sequence?* A different question is *interventional*: given a reference genomic window and an explicit edit, what is the *consequence* of that edit in representation space? If a frozen encoder’s response to an edit could be *anticipated*—computed from the reference state plus a description of the edit, without re-running the encoder—then a static genomic encoder would become an action-conditioned *world model* [10, 11] whose actions are discrete genomic edits, and whose latent transitions are cheap enough to score variants, roll out haplotypes, and plan edit sets. That is the promise: one expensive encoder pass per reference window, then arbitrarily many cheap latent queries (fig. 1).

This paper measures whether that promise can be kept, and reports that it cannot—for a reason more general than any implementation. Our object of study is the *edit-response geometry* of a frozen DNA foundation model, principally Carbon-500M [13]: the displacement $\Delta = s_{\text{alt}} - s_{\text{ref}}$ induced in a pooled representation by a single-nucleotide edit. We measure it on 14,000 GRCh38-grounded variants across four pooling widths, and we ask three questions of it in sequence. Does the frozen encoder *register* the edit informatively? Is that a property of genomic foundation models, or of Carbon? And can the registration be *predicted* from the pooled reference state and the action?

The answers dissociate, and the dissociation is the result. *On Carbon*, the edit is registered richly: Δ separates ClinVar pathogenic from benign SNVs at AUROC 0.9259, and this survives every control we could construct—restriction to SNVs, restriction to within a substitution class, conditioning on the full trinucleotide context, and probes that see the reference locus but never the edit. That richness, however, is *Carbon’s* and not the field’s: run through two encoders sharing neither objective nor tokenizer, the identical measurement on the identical variants gives 0.7574 on Nucleotide Transformer v2 and 0.5460—near chance—on HyenaDNA (section 5). And on Carbon, where the response is worth predicting at all, it cannot be anticipated. The best learned predictor of Δ from (s_{ref}, a) recovers AUROC 0.6520, against a ceiling of 0.9259 from reading the encoder’s true response, and against 0.6306 available from the substitution class alone with no model at all. Adding the pooled reference state to the action buys approximately nothing—and under a linear model it is worse than nothing, scoring 0.5854 against 0.6079 for a lookup table that ignores the reference state entirely. **The information exists; it is not where a latent world model needs it to be. There is no shortcut through the pooled latent: you must run the encoder on the edited sequence.**

The predict-no-change trap, which is universal. A second measurement explains why this failure mode is easy to misread, and why we consider the negative result worth reporting rather than quietly abandoning. At global pooling, $\cos(s_{\text{ref}}, s_{\text{alt}}) = 0.99989$: the edited state is cosine-indistinguishable from the reference to four decimal places. Yet AUROC over those same states is 0.9313. An objective that asks a model to predict s_{alt} from s_{ref} —which is what the natural JEPA-style formulation asks—is therefore *degenerate*: “predict no change” attains the target to within 10^{-4} while carrying none of the discriminative signal. Dilution of the edit under pooling is real and strong ($9.4\times$ in relative displacement), but it is *not* information loss. The trap is the metric, not the representation. An experiment that reports a cosine gap under this formulation has measured its own objective, not the encoder. Unlike the informativeness result, this one *does* generalise: across all three encoders and every pooling width, $\cos(s_{\text{ref}}, s_{\text{alt}}) \geq 0.993$, reaching 1.00000 for HyenaDNA at global pooling (section 5). It is a property of pooling a genomic FM, not of Carbon.

This is a measurement paper with a negative headline. We did not build a working genomic world model, and we do not claim one. We report what a frozen encoder’s pooled edit response contains, what it costs to obtain, and what cannot be learned from it. The intended audience is other groups considering the same architecture; the intended saving is their compute. We are also not proposing a variant-effect predictor: section 7 shows the measurement is strong on curated binary labels and *weak* on quantitative function (Spearman -0.265 on BRCA2 saturation genome editing), and we report that as a limitation rather than an aside.

Contributions.

1. **An edit-response spectroscopy of Carbon-500M** (section 4): the displacement induced by a single-base edit, measured across pooling widths on 14,000 GRCh38-grounded variants, with the

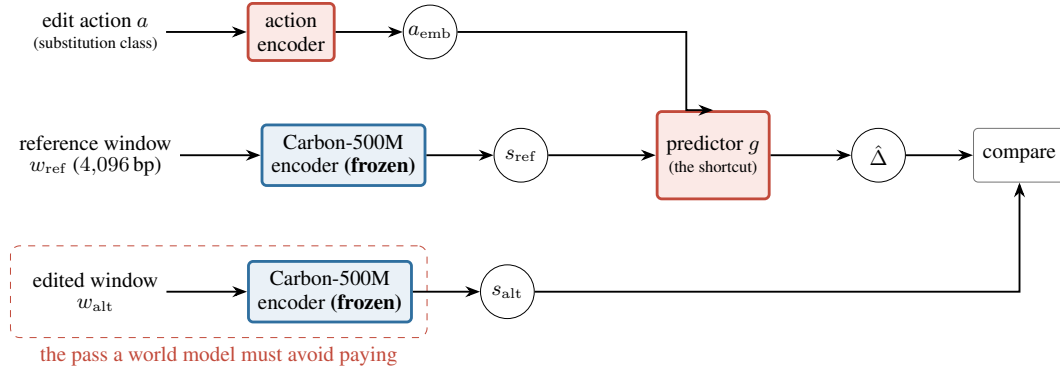


Figure 1: **The object of study.** A latent world model over a frozen genomic encoder freezes Carbon-500M, encodes a reference window once to a pooled state $s_{\text{ref}} \in \mathbb{R}^{1024}$, and trains a small predictor g to anticipate the effect of an edit action a directly in latent space. Its entire value proposition is that the lower (red, dashed) path—running the encoder on the edited window—never has to be taken. We measure the true displacement $\Delta = s_{\text{alt}} - s_{\text{ref}}$ on both paths and ask whether $g(s_{\text{ref}}, a)$ can reproduce it. Section 6 reports that it cannot: the discarded pass is where the pathogenicity signal is created. Section 8 describes the released v0.2.1-r1 instantiation of this diagram and the implementation defects that made that particular run uninterpretable.

sensitivity ceiling ($\cos = 0.99989$ at global pooling) and the dissociation between that ceiling and a flat 0.92–0.93 AUROC.

2. **A controlled demonstration that Carbon-500M’s response is interventional** (section 4.2): the signal is not the mutation spectrum (0.6306 with no geometry), not the exact local sequence context (conditioning on the full trinucleotide context costs 0.0059 AUROC), and not the genomic neighbourhood (0.6392 from a reference-only probe, tracked by a chromosome-held-out probe at 0.6354, so not locus leakage).
3. **A propagation measurement** (section 4.3): the local and distal components of the response point in genuinely different directions ($\cos = 0.4714$), so pooling does not merely rescale a local token swap; Carbon propagates the edit.
4. **A three-encoder generality test that splits the claim in two** (section 5): the same 12,993 SNVs through Carbon, Nucleotide Transformer v2, and HyenaDNA show that the predict-no-change trap is *universal* ($\cos \geq 0.993$ everywhere), while informativeness is *not* ($0.93 \rightarrow 0.76 \rightarrow 0.55$) and the interventional property is Carbon-specific—on Nucleotide Transformer v2 at ± 48 bp the unedited window predicts pathogenicity better than the edit response does. This is why we scope the interventional claim to Carbon throughout.
5. **The core negative result** (section 6): the pathogenicity-bearing component of the edit response is precisely the component that is *not* predictable from the pooled reference state plus the action, with the failure worsening as the state is pooled more—as an information-bottleneck account predicts.
6. **An explained retraction** (section 8): our own withdrawn v0.2.1-r1 release had real implementation defects, but we show the deeper reason it could not have worked, and that a correct implementation would have failed too.

What we do not claim. No clinical, diagnostic, or decision-support claim of any kind. No claim that this is a competitive variant-effect predictor—and, as section 7.2 shows, our ClinVar slice is an easy one, so the AUROCs here are cohort-dependent and are not comparable to published benchmark numbers. No claim about token-level world models, which we did not test. No claim that Carbon’s informative, interventional edit response generalises to genomic foundation models at large: section 5 shows it does not, which is why every interventional statement in this paper is scoped to Carbon by name. No claim about encoders larger than 500M or with longer contexts than we tested, and in particular no claim to know whether the spread we measure tracks architecture, objective, or capacity. We collect these boundaries in section 9.

2 Related Work

Latent world models over frozen features. The template we test is DINO-WM [27], which learns an action-conditioned latent forward model on top of frozen DINOv2 [22] patch features and plans in latent space by predicting future patch embeddings. Sobal et al. [26] isolate what a latent dynamics model buys over model-free alternatives on reward-free offline data; the Dreamer line [11] and TD-MPC2 [12] establish, respectively, generative and decoder-free latent dynamics for control; Ha and Schmidhuber [10] originate the frozen-encoder plus latent-controller decomposition. Our setting differs in a way that turns out to be decisive. These are *high action-authority* domains: an action moves the state substantially, so “predict-no-change” is a weak baseline and the target signal is rich. A single-nucleotide edit moves a pooled genomic state by $\cos = 0.99989$ at global pooling, so predict-no-change is nearly unbeatable in absolute-state space, and the science has to move into displacement space (section 4). We are not planning; we are measuring the ceiling that any such world model would be built against.

JEPAs and anti-collapse regularization. I-JEPA [1] predicts target-block representations from a context block without a pixel decoder, and V-JEPA 2 [2] adds an action-conditioned variant enabling planning from image goals—the closest visual analogue of the predictor we test, with an SNV as the action. LeJEPA [4], by Balestriero and LeCun, proves that among embedding distributions at a fixed covariance budget the isotropic Gaussian uniquely minimizes downstream probe bias, and enforces it with *SIGReg*: random one-dimensional projections with characteristic-function matching, in linear time and memory. **We adopt LeJEPA’s isotropic-Gaussian rationale, not its estimator.** The regularizer in our own withdrawn release (section 8) was a closed-form Gaussian KL to an isotropic target—a different and weaker mechanism than *SIGReg*, which we describe here only so that run is characterized accurately. VICReg, Barlow Twins, and DINO supply the surrounding anti-collapse toolbox. The difference from all of these is one of regime rather than method: in vision, the masked target is high-uncertainty and information-rich; here the target is $\approx 99.99\%$ recoverable by copying the input, and the entire signal lives in the residual.

Genomic foundation models and their variant-effect paradigm. Carbon is autoregressive and decoder-only with a 6 bp tokenizer, so, like Evo and Evo 2 [7, 21], GPN [5], and HyenaDNA [20], it supports zero-shot Δ log-likelihood scoring; masked models such as the Nucleotide Transformer [9], DNABERT [14], and Caduceus [25] support pseudo-likelihood analogues. AlphaGenome [3] and supervised track predictors are a contrast class: they score variants by comparing predicted assay tracks, not by a likelihood or a state displacement. Carbon itself has no archival publication; we cite the technical report and pin a model revision (table 9). Every one of these papers reports downstream VEP *accuracy*; none characterizes the intrinsic displacement geometry of a single edit, which is our object. We measure that geometry on three of these models—Carbon, Nucleotide Transformer v2, and HyenaDNA—and one of our findings is that it differs sharply between them (section 5).

Variant-effect prediction: the Δ log-likelihood and embedding-distance paradigms. ESM-1v [19] established that a frozen LM’s log-likelihood ratio predicts mutation effect with no task supervision, and GPN [5] and Evo [21] carry the identical idea into DNA; AlphaMissense [8] gives broader field context. Our nearest methodological neighbour is **embedding-distance VEP**: BEND [18] and general in-silico-mutagenesis practice already use the distance between reference and alternate embeddings as a variant score. That is precisely the quantity whose geometry we dissect. Prior work *thresholds* $\|\Delta\|$ as a black-box score; we characterize why it behaves as it does and—the part that is new—ask whether it can be *anticipated* rather than measured. The evaluation resources are ClinVar [16], gnomAD [15], TraitGym [6], and MaveDB [23], specifically the BRCA2 saturation genome-editing assay of Sahu et al. [24].

The gap, and the closest concurrent work. Across this literature we find no systematic characterization of the pooled-state displacement geometry of a single-nucleotide edit, and no test of whether that displacement is recoverable by an action-conditioned latent predictor. The most direct concurrent neighbour is JEPA-DNA [17], which grounds genomic FMs by predicting masked-segment latent representations and improves linear-probe accuracy. It is a *pretraining objective evaluated by downstream accuracy*; ours is a *measurement of single-edit displacement geometry and its predictability*. The two are complementary, and—if our bottleneck account is right—JEPA-DNA’s

choice to predict *segment* rather than pooled representations is on the more promising side of exactly the boundary we draw in section 9.

3 Method

3.1 Frozen encoder and the representation contract

The encoder is Carbon-500M [13], a 500M-parameter decoder-only DNA language model (LLaMA-style, hidden size 1024, 6 bp tokenizer), loaded at pinned revision 5d31d59b in bf16 and **frozen throughout**. No encoder parameter is trained anywhere in this paper. Each variant is placed at the centre of a 4,096 bp GRCh38 window. We take layer-20 hidden states $H \in \mathbb{R}^{T \times 1024}$ and pool them to a state $s \in \mathbb{R}^{1024}$.

Three contracts fix the measurement. We state them explicitly because the failure documented in section 8 was caused by leaving them implicit.

Window. Reference and edited windows are the same 4,096 bp interval, differing in exactly one base for an SNV. The variant is at the window centre by construction.

Pooling. $s^{(r)} = \text{mean}(H[c-r : c+r+1])$ for radius $r \in \{8, 64, 256\}$ tokens, where c is the token index of the edited base, derived from the observed DNA-content offset after Carbon’s leading control token. Because the tokenizer is 6 bp, these radii are $\pm 48, \pm 384$, and ± 1536 bp. We write $r=0$ for global-mean pooling over the entire window, which is therefore the *widest* setting, not the narrowest: throughout, the pooling grid is ordered $r=8 < r=64 < r=256 < \text{global}$ by width. Reference and edited states are always pooled at the same centre with the same radius.

Normalization. All geometry is defined on raw pooled states. We report only quantities that are explicitly scale-aware ($\|\Delta\|_2$) or explicitly scale-free ($\cos, \|\Delta\|_2 / \|s_{\text{ref}}\|_2$), so that no comparison depends on an unstated normalization convention.

3.2 Definitions

For a variant v with reference window w_{ref} and edited window $w_{\text{alt}} = \text{apply}(v, w_{\text{ref}})$, at pooling width r :

$$s_{\text{ref}} = \text{enc}(w_{\text{ref}}), \quad s_{\text{alt}} = \text{enc}(w_{\text{alt}}), \quad \Delta = s_{\text{alt}} - s_{\text{ref}}, \quad (1)$$

and we report the *relative displacement* and the *state cosine*

$$\text{rel}\Delta = \frac{\|\Delta\|_2}{\|s_{\text{ref}}\|_2}, \quad \cos(s_{\text{ref}}, s_{\text{alt}}) = \frac{\langle s_{\text{ref}}, s_{\text{alt}} \rangle}{\|s_{\text{ref}}\|_2 \|s_{\text{alt}}\|_2}. \quad (2)$$

$\text{rel}\Delta$ measures how far the edit moves the state relative to the state’s own scale; $\cos(s_{\text{ref}}, s_{\text{alt}})$ measures how far it moves it in the metric a JEPA-style objective would actually optimize. Section 4 shows these two tell the same story, and that both are misleading about information content.

3.3 Cohort

The cohort is 14,000 real GRCh38-grounded variants $\times$ 4 pool radii = 56,000 rows, with 0 skips: ClinVar [16] 1,572 pathogenic / 6,428 benign, and 6,000 BRCA2 saturation genome-editing variants [24] from MaveDB score set urn:mavedb:00001242-a-1 [23]. Because the initial ClinVar set was not SNV-only, and because indels displace the state far more than substitutions in a way that is aligned with the label, all headline analyses restrict to SNVs: 12,993 SNVs, of which ClinVar contributes 899 pathogenic and 6,094 benign, with 1,007 non-SNV rows excluded and counted. Every one of the 12,993 SNVs was checked against the reference FASTA and matched its recorded reference base (12,993/12,993).

3.4 Two further encoders, and the units that make them comparable

Everything above describes one encoder, and a measurement on one encoder cannot distinguish a property of pooled genomic FM states from a property of Carbon. To separate them (section 5) we repeat the identical measurement on two further frozen encoders, chosen to share as little as possible with Carbon or with each other:

Nucleotide Transformer v2, 100M multi-species [9]: a *masked* language model—a different training objective—with a 6 bp tokenizer and $d=512$.

HyenaDNA-450k [20]: implicit long convolutions rather than attention, and a *single-nucleotide* (character) tokenizer, with $d=256$.

Three objectives, two tokenizations, three widths. Both additional encoders are frozen at load, both see the same 12,993 SNVs in the same 4,096 bp windows at the same four pooling widths—51,972 rows each, 0 skips—and both are processed by `tools/research/edit_response_analysis.py` *unchanged*, so no cross-encoder difference we report can come from the analysis.

Radii are tokens; every comparison is in base pairs. The pool radius r is counted in *tokens*. Carbon and NT-v2 pack 6 bp into a token; HyenaDNA packs 1. Comparing the three at equal r would therefore compare a ± 48 bp pool against an ± 8 bp one and call the difference an encoder effect. We instead run HyenaDNA at radii 0, 48, 384, 1536 so that its pooling windows match the others’ 0, 8, 64, 256 at 0 (global), ± 48 , ± 384 and ± 1536 bp, and we label every cross-encoder axis, row, and column **in base pairs**. The distinction is not cosmetic: a single base perturbs 1 of 4,097 character tokens against 1 of 683 6 bp tokens, which is the mechanism behind the encoder differences in table 2.

A tokenizer hazard worth recording: NT-v2 re-anchors its grid on N. NT-v2’s tokenizer packs 6-mers greedily behind a leading `<cls>`, which makes the obvious mapping from a genomic offset to a token index— $\lfloor \text{locus}/6 \rfloor + 1$ —look safe. It is not. A 6-mer containing an N is out of vocabulary; the tokenizer degrades to single-nucleotide tokens across it and *re-anchors the 6-mer grid after it*. Any window containing an N—and reference windows do—therefore has a token layout that the arithmetic mapping silently gets wrong, and the pool is centred on the wrong tokens with no error raised anywhere. Our tool instead reconstructs each token’s span from the token strings themselves, verifies that the reconstructed spans tile the window exactly, and fails closed on any `<unk>`. We record this because the failure is silent and label-independent: it is precisely the class of defect that survives review, and it is a close cousin of the pooling-coordinate defect that withdrew our own earlier release (section 8).

3.5 Predictability protocol and its baselines

The world-model question, correctly posed, is not “predict s_{alt} from s_{ref} ”—section 4 shows that target is $\approx 99.99\%$ attained by copying the input. It is: *predict the displacement Δ from the reference state and the action, without encoding the edited window*. That is exactly the efficiency thesis, and it is falsifiable.

The action space is the 12 substitution classes (ordered base-to-base pairs); position is not an action variable because every variant sits at the window centre by construction. We fit predictors $g(s_{\text{ref}}, a) \rightarrow \hat{\Delta}$ under 5-fold cross-validation on the 12,993 SNVs, at radius 8—the most favourable, most local pooling width, chosen deliberately so that a failure there is informative about every wider setting. We evaluate two things, and keeping them separate is essential:

Direction: $\cos(\hat{\Delta}, \Delta)$, the fidelity of the predicted displacement direction.

Magnitude: AUROC of $\|\hat{\Delta}\|_2$ for ClinVar pathogenic vs. benign—whether the predicted response carries the pathogenicity signal that the *true* response carries.

The baseline ladder. A negative result is only as good as what it is negative against, so we bracket the learned predictors from below and above. From below: a *global-mean* Δ predictor that ignores its inputs entirely; a *per-substitution-class-mean lookup*, which is a 12-entry table with no learning; and *the substitution class alone*, scored directly with no displacement at all. From above: the *TRUE* Δ ceiling—what you get by paying for the encoder pass. A world model is worth building only if it lands meaningfully above the lookup table and near the ceiling.

Three method notes that constrain interpretation. First, a **cosine-trained predictor’s magnitude is meaningless**: the cosine objective is scale-invariant, so $\|\hat{\Delta}\|$ is unconstrained, and any AUROC computed from it is an artifact of initialization and optimizer path rather than a measurement.

We therefore report $\text{AUROC}(\|\hat{\Delta}\|)$ *only* for the MSE-trained predictor, and report the cosine-trained predictor’s direction only.

Second, the ridge baseline’s λ was swept over $1\text{--}10^6$ (best 10^3) before we concluded anything: an under-regularised failure would prove nothing. Ridge is fit with an intercept. This detail is load-bearing rather than pedantic: an earlier prototype standardised a design matrix containing a constant column, and because that column has zero standard deviation, standardising annihilated it, leaving the model unable to represent $\mathbb{E}[\Delta]$. Every ridge number reported here comes from the corrected fit.

Third, **AUROC is computed with averaged tied ranks**. This matters only for the model-free baselines and it matters a great deal there: the global-mean predictor emits 5 distinct magnitudes and the class-mean lookup 60, so ties dominate their score and a tie-naïve implementation misreports them. The MLP-MSE predictor emits 12,993 distinct values with no ties and is unaffected either way—which is why the headline is robust to this choice while the baselines are not. All predictability numbers are those emitted by `tools/research/delta_predictability.py`, using the repository’s tie-safe AUROC.

4 Edit-Response Spectroscopy of Carbon-500M

This section measures Carbon-500M alone; section 5 then asks which of its findings are properties of Carbon and which are properties of pooling a genomic FM. The answer is not uniform, so we name the encoder rather than the field throughout.

4.1 The sensitivity ceiling: dilution is real, information loss is not

Figure 2 sweeps the pooling width. Sensitivity peaks at the narrowest pool and decays monotonically as the pool widens: mean $\text{rel}\Delta$ is 0.0759 at $r=8$, 0.0230 at $r=64$, 0.0101 at $r=256$, and 0.0083 at global pooling—a $9.4\times$ swing. The state cosine mirrors it: $\cos(s_{\text{ref}}, s_{\text{alt}})$ is 0.99352 at $r=8$, 0.99925 at $r=64$, 0.99984 at $r=256$, and 0.99989 globally. Under global pooling a single-base edit is invisible to cosine at four decimal places.

The natural inference is that pooling has destroyed the edit. It has not. Over the very same states, ClinVar pathogenic-vs-benign AUROC is *flat*: 0.9259 at $r=8$, 0.9278 at $r=64$, 0.9313 at $r=256$, and 0.9313 globally (table 4). The edit signal survives at every pooling scale. What decays is not the information but its *share of the norm*.

This dissociation is the paper’s first substantive point, and it retro-diagnoses a whole class of experiments including our own. **“Predict-no-change” wins on cosine because cosine is the wrong metric, not because the information is absent**. A predictor optimizing $1 - \cos(\hat{s}_{\text{alt}}, s_{\text{alt}})$ against a target that the identity function already reaches at 0.99989 is being scored almost entirely on a component of the problem that carries no signal. The objective is degenerate by construction. Displacement space is not a stylistic preference here; it is the only space in which the question is well posed.

Of everything in this section, this is the finding that generalises: section 5 shows the same ceiling on a masked model and on a convolutional, character-level one, with $\cos(s_{\text{ref}}, s_{\text{alt}}) \geq 0.993$ at every encoder and every pooling width.

4.2 Carbon-500M’s response is interventional, not a confound

An AUROC of 0.93 from a frozen encoder invites suspicion, so we tried to break it (table 4, fig. 3). Everything in this subsection is a statement about Carbon: section 5 shows that the interventional conclusion we reach here is one Carbon earns and the other two encoders we measured do not.

Not indels. The raw cohort was not SNV-only, and indels displace the state far more than substitutions in a way that correlates with the pathogenic label—an inflation aligned with the outcome. Purging them (1,007 non-SNV rows, counted and excluded) leaves AUROC at 0.9259 at $r=8$. The signal was not the indels.

Not the mutation spectrum. Some substitution classes are simply more often pathogenic, and a model could exploit that without any geometry at all. Scoring by substitution class alone—no

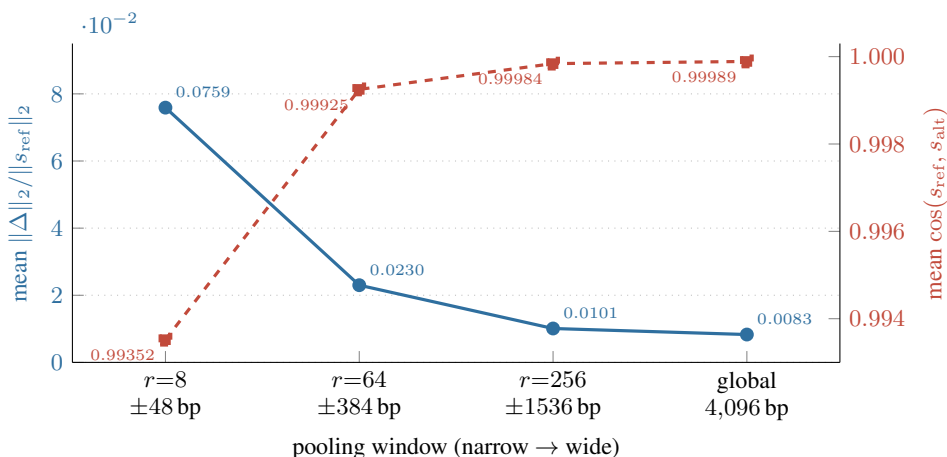


Figure 2: The sensitivity ceiling: a single-base edit is nearly invisible to cosine at every pooling width, and essentially invisible at global pooling. Left axis (blue, solid): mean relative displacement $\|\Delta\|_2 / \|s_{\text{ref}}\|_2$. Right axis (red, dashed): mean $\cos(s_{\text{ref}}, s_{\text{alt}})$. Both are means over the 14,000-variant cohort. Sensitivity peaks at the narrowest pool ($r=8$, ± 48 bp) and falls 9.4× as the pool widens to the full window, where the reference and edited states are cosine-identical to four decimal places (0.99989). *This is not information loss.* Pathogenicity AUROC over the same states is flat at 0.92–0.93 across this entire sweep (fig. 3). A world model trained to predict s_{alt} from s_{ref} under a cosine or absolute-state objective is therefore optimising against a target that “predict-no-change” already attains to within 10^{-4} : the objective is degenerate because the metric is wrong, not because the signal is absent.

displacement, no model—gives 0.6306. Restricting the AUROC to comparisons *within* each of the 12 substitution classes (covering $n=6,993$) leaves 0.9202–0.9245 across the sweep. The geometry carries far more than the spectrum, and it carries it after the spectrum is held fixed.

Not the local sequence context either—the sharpest control. The 12-class control is weaker than it looks, because it does not condition on local sequence, and a DNA language model’s response to a base swap plausibly depends on its immediate neighbours. So we ran the standard mutational-signature decomposition (table 1): condition on the full trinucleotide context—the 5’ base, the reference base, the 3’ base—and the substitution, then score only within each context. This is the control a referee is most likely to demand, and it is the one that could most plausibly have destroyed the result. It does not. Conditioning on the exact local context costs 0.0059 AUROC: 0.9305 stratified against 0.9364 unstratified, both measured on the identical 5,027 variants (745 pathogenic) covered by the 37 contexts with at least 10 variants per class. Meanwhile the trinucleotide lookup with zero geometry reaches 0.6618—above the 12-class lookup’s 0.6306, as strictly more conditioning information should be, and still far below the geometry. The edit response is not a restatement of mutation chemistry or of local k -mer identity.

The apples-to-apples framing is load-bearing and we want to be explicit about it. The stratified 0.9305 exceeds the full-cohort unstratified 0.9259 of table 4, which would look like stratification *improving* the signal. It does not: the 37 retained contexts cover a different and slightly easier subset of the cohort, and against its own subset the stratified score is a small *decrease*. We report the comparison within the shared population because it is the only one that licenses a conclusion.

We flag the limit of this control rather than let it be inferred: trinucleotide context spans ± 1 bp, whereas Carbon’s tokenizer packs 6 bp into a token. It therefore does not fully control for the exact ref-6-mer $\rightarrow$ alt-6-mer token pair. The natural next stratification, ± 2 bp (5-mers), is too sparse to be powered at 899 pathogenic variants, so we do not report it (section 9).

Not the genomic neighbourhood. This was the most serious threat: pathogenic variants cluster in conserved, functionally dense regions, so a probe might recognise the *locus* and never need the edit. A linear probe on s_{ref} alone—which never observes the edit—reaches 0.6392 at $r=8$, and 0.5500–0.5741 at the wider settings. To exclude locus leakage across cross-validation folds, we

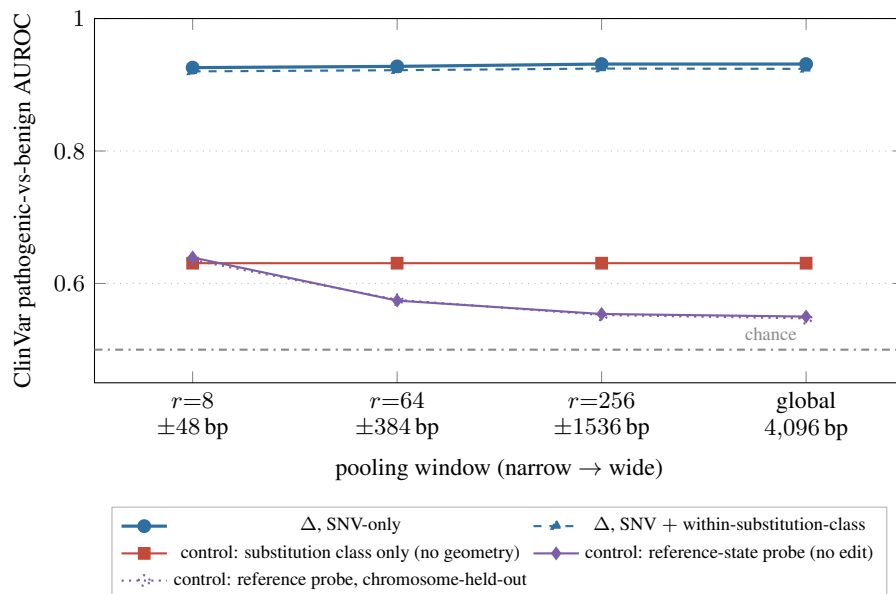


Figure 3: The edit response is informative and interventional, and it is flat in the pooling width. The displacement Δ separates ClinVar pathogenic from benign SNVs at AUROC 0.92–0.93 at every pooling width, including global pooling where $\cos(s_{\text{ref}}, s_{\text{alt}}) = 0.99989$ (fig. 2). Three controls bound the trivial explanations. Scoring variants by *substitution class alone*, with no geometry and no model, reaches 0.6306; restricting the comparison to *within* a substitution class leaves the signal essentially intact (0.9202–0.9245), so the signal is not the mutation spectrum. A linear probe on the reference state alone—which never sees the edit—reaches at most 0.6392, and a chromosome-held-out probe tracks it (0.6354), so the signal is not the genomic neighbourhood and is not locus leakage. What is left is the edit. All numbers in table 4; the sharper trinucleotide-context control is in table 1.

re-ran the probe with chromosome-disjoint folds: 0.6354 at $r=8$, tracking the random-fold probe at every radius. There is no same-locus leakage, and the neighbourhood explains at most 0.64 of the 0.93.

What remains, after the spectrum and the neighbourhood are removed and leakage is excluded, is the edit itself. Carbon-500M’s response is *interventional*. We are deliberate about the subject of that sentence: run the same control on Nucleotide Transformer v2 at the same span and it inverts, the reference-only probe beating the edit response (0.7794 against 0.7496; section 5). The control is passed by an encoder, not by a field.

4.3 Carbon propagates the edit

If a single-base edit merely rewrote its own 6 bp token, then the displacement at a wide pool would be a scaled-down copy of the displacement at a narrow one, and the whole phenomenon would be an uninteresting artifact of averaging. We test this directly by comparing displacement *directions* across pooling widths (fig. 4). Per variant, $\cos(\Delta \text{ at global pooling}, \Delta \text{ at } r=8)$ has mean 0.4714, median 0.4721, and standard deviation 0.1905; only a fraction 0.004 of variants exceed 0.9, and none exceed 0.99. A pure local token swap would put this distribution at 1.0. It sits near 0.47.

Magnitude *ordering*, by contrast, is well preserved: Spearman between $\|\Delta\|$ at global pooling and at $r=8$ is 0.8595. The picture is coherent: *which* variants move the state a lot is stable across scales—which is why AUROC is flat—but *where* they move it is not. The distal response is a genuinely different direction from the local one. This is an attention-mediated, context-dependent response, and the local and distal components independently carry the signal. It also rules out the remaining deflationary reading of section 4.1: the flat AUROC under wide pooling is not a local vector surviving in miniature, it is a distinct global response.

Table 1: **The trinucleotide-context control: conditioning on the exact local sequence costs essentially nothing.** Radius 8, ClinVar SNVs. This is the standard mutational-signature decomposition: we condition on the 5' base, the reference base, the 3' base, *and* the substitution, then score only *within* each trinucleotide context. If the edit response were a restatement of mutation chemistry or of local k -mer identity, stratifying it away would collapse the AUROC. It costs 0.0059. The comparison is made *apples-to-apples*, on the identical 5,027 variants that the 37 sufficiently populated contexts cover: comparing the stratified number against the full-cohort 0.9259 would compare different populations and overstate the result. The two zero-geometry lookups are reported on the full cohort for reference; the trinucleotide lookup beats the 12-class lookup, as strictly more conditioning information should, and both remain far below the geometry. *Caveat:* trinucleotide context spans ± 1 bp while Carbon’s tokenizer packs 6 bp per token, so this does not fully control the exact ref-6-mer $\rightarrow$ alt-6-mer token pair (section 9).

Scoring	n	AUROC $\uparrow$
<i>Apples-to-apples, on the same subset (745 pathogenic)</i>		
Unstratified $\ \Delta\ $	5,027	0.9364
Within-trinucleotide-context pooled	5,027	0.9305
<i>cost of conditioning on exact local context</i>		−0.0059
<i>Full-cohort references</i>		
Unstratified $\ \Delta\ $	6,993	0.9259
Trinucleotide-context lookup, zero geometry	6,993	0.6618
Substitution-class lookup (12 classes), zero geometry	6,993	0.6306
37 trinucleotide contexts retained (≥ 10 variants per class).		

5 Generality: The Trap Is Universal, the Signal Is Not

Everything in section 4 is a measurement of one encoder, and two quite different claims are entangled in it. One is that a single edit barely moves a pooled genomic state, so any absolute-state objective is degenerate—a claim about *pooling*, which ought to hold for any encoder. The other is that the displacement is informative and interventional—a claim we have so far only licensed for *Carbon*. A paper that leaves these fused generalises an $n=1$ result by accident. So we ran the identical measurement, on the identical 12,993 SNVs, through two encoders that share neither training objective nor tokenizer with Carbon or with each other (section 3.4), with the analysis code unchanged. The two claims come apart, and the second one does not survive.

5.1 The sensitivity ceiling replicates everywhere

Table 2 reports the ceiling at bp-matched spans. It holds on all three encoders. The lowest state cosine anywhere in the table is Carbon’s 0.99352 at ± 48 bp, and since ± 48 bp is the *maximum*-sensitivity span on all three, $\cos(s_{\text{ref}}, s_{\text{alt}}) \geq 0.993$ at every encoder and every pooling width. Under global pooling the edit is invisible to four decimal places on Carbon (0.99989) and on NT-v2 (0.99998), and to five on HyenaDNA, whose cosine is 1.00000. The dilution curve has the same shape everywhere: sensitivity is highest at the narrowest pool and falls as the pool widens (rel Δ of 0.0759 $\rightarrow$ 0.0083 on Carbon, 0.0889 $\rightarrow$ 0.0050 on NT-v2, 0.0130 $\rightarrow$ 0.0008 on HyenaDNA).

HyenaDNA is the extreme case, and the instructive one: its response is the smallest by roughly an order of magnitude (rel $\Delta = 0.0008$ globally against Carbon’s 0.0083; 0.0130 against 0.0759 at ± 48 bp). Character tokenization predicts exactly this: a single base perturbs 1 of 4,097 tokens rather than 1 of 683.

This strengthens the paper’s first claim rather than qualifying it. The predict-no-change trap is not an artifact of Carbon’s tokenizer, its autoregressive objective, or its width. Across an autoregressive model, a masked model, and a convolutional one, an objective that scores a predicted edited state against the true one by cosine is being scored on a target the identity function already attains to at least three decimal places. Anyone building an action-conditioned latent predictor over pooled DNA-LM features will hit this, whichever encoder they pick.

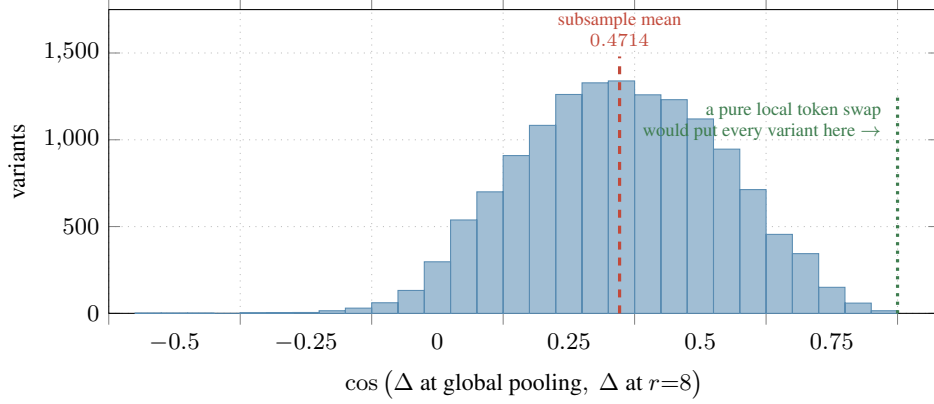


Figure 4: **Carbon propagates the edit; wide pooling is not a rescaling of a local vector.** Per-variant cosine between the displacement measured at global pooling and the displacement measured at $r=8$ (± 48 bp). If a single-base edit merely rewrote its own token, and wider pooling only shrank that same vector, this distribution would be a spike at 1.0. It is not: on the verified $n=2,000$ subsample the mean is 0.4714, the median 0.4721, and the standard deviation 0.1905; a fraction of only 0.004 of variants exceeds 0.9 and none exceeds 0.99. The distal response is a genuinely different direction from the local one—an attention-mediated, context-dependent response rather than a token substitution. Magnitude *ordering*, by contrast, is preserved across pooling widths (Spearman 0.8595 between $\|\Delta\|$ at global pooling and at $r=8$), which is why AUROC is flat in fig. 3 while direction is not conserved. Histogram over all 14,000 variants; the quoted statistics are the verified subsample values.

Table 2: **The predict-no-change trap replicates on every encoder we measured.** The same 12,993 SNVs in the same 4,096 bp windows, through three frozen encoders that share neither training objective nor tokenizer, at **bp-matched** pooling spans (section 3.4). A single-base edit leaves the pooled state cosine-indistinguishable from the reference on all three: the smallest cosine anywhere in this table is Carbon’s 0.99352, and ± 48 bp is the *maximum*-sensitivity span, so every intermediate span is closer to 1 still. HyenaDNA is the extreme case— $\cos = 1.00000$ under global pooling—and the mechanism is visible in the tokenizer column: one base perturbs 1 of 4,097 character tokens against 1 of 683 6 bp tokens, and its relative displacement is correspondingly $\approx 10\times$ smaller than Carbon’s at global pooling and $\approx 6\times$ smaller at ± 48 bp. The dilution curve has the same shape everywhere: sensitivity peaks at the narrowest pool and falls as the pool widens. Any objective scored by $\cos(s_{\text{ref}}, s_{\text{alt}})$ is degenerate on all three encoders. Whether the displacement is *informative*, however, does not replicate at all (table 3).

Encoder	Objective / tokenizer / width	$\cos(s_{\text{ref}}, s_{\text{alt}})$		$\text{rel}\Delta = \ \Delta\ _2 / \ s_{\text{ref}}\ _2$	
		global	± 48 bp	global	± 48 bp
Carbon-500M	autoregressive, 6 bp token, $d=1024$	0.99989	0.99352	0.0083	0.0759
NT-v2-100M	masked LM, 6 bp token, $d=512$	0.99998	0.99479	0.0050	0.0889
HyenaDNA-450k	implicit conv., 1 bp token, $d=256$	1.00000	0.99991	0.0008	0.0130

5.2 Informativeness does not replicate, and the interventional property is Carbon’s

The second claim fails, and it fails in a graded way (table 3, fig. 5). The same $\|\Delta\|$, on the same variants, from the same code, separates ClinVar pathogenic from benign at 0.9313 on Carbon, 0.7574 on NT-v2, and 0.5460—near chance—on HyenaDNA, under global pooling; the ordering is stable across spans. Whether a frozen genomic FM’s pooled state registers a point edit *usefully* is therefore not a property of genomic FMs. It is a property of the encoder.

The sharper version concerns the control that section 4.2 rests on. The fair test of whether the *edit* beats the *neighbourhood* is a supervised probe on Δ against the same supervised probe on s_{ref} , which never observes the edit; we write the difference Δ adds. Carbon clears it at every span, by $+0.1346$ at ± 48 bp, $+0.2967$ at ± 1536 bp, and $+0.3075$ globally. HyenaDNA adds $+0.0229$ to

Table 3: **Informativeness is encoder-specific, and the interventional property is *Carbon*-specific.** Same 12,993 SNVs, same windows, same analysis code, bp-matched spans (section 3.4). $\|\Delta\|$ is the unsupervised displacement magnitude; $\text{dir}(\Delta)$ is a supervised 5-fold directional probe on Δ ; *Ref. probe* is the same supervised probe on s_{ref} alone, which never observes the edit. Δ *adds* is $\text{dir}(\Delta)$ minus the reference probe—probe against probe, so it is the fair test of whether the *edit* beats the *neighbourhood*, and it is the quantity that separates the three encoders. Carbon clears it at every span. HyenaDNA adds ≈ 0.02 from a base of chance. Nucleotide Transformer v2 goes *negative* at ± 48 bp: its unedited window predicts pathogenicity better than its edit response does. The BRCA2 column is Spearman($\|\Delta\|$, function) at ± 48 bp, and is not reported for HyenaDNA.

Encoder	Span	$\ \Delta\ $	$\text{dir}(\Delta)$	Ref. probe	Δ adds
Carbon-500M	± 48 bp	0.9259	0.7737	0.6392	+0.1346
	± 1536 bp	0.9313	0.8506	0.5539	+0.2967
	global (4,096 bp)	0.9313	0.8575	0.5500	+0.3075
NT-v2-100M	± 48 bp	0.7616	0.7496	0.7794	−0.0298
	± 1536 bp	0.7590	0.7232	0.5907	+0.1326
	global (4,096 bp)	0.7574	0.7119	0.5823	+0.1296
HyenaDNA-450k	± 48 bp	0.5676	0.5795	0.5566	+0.0229
	± 1536 bp	0.5460	0.5814	0.5545	+0.0269
	global (4,096 bp)	0.5460	0.5824	0.5521	+0.0303

BRCA2 sGE, Spearman($\|\Delta\|$, function) at ± 48 bp ($n=6,000$): Carbon -0.265 , NT-v2 -0.186 .

+0.0303—nothing, and from a base of chance. NT-v2 adds +0.1296 and +0.1326 at the two wide spans but **−0.0298** at ± 48 bp, where its *unedited* window (0.7794) predicts pathogenicity better than its edit response does (0.7496). At that span, on that encoder, the interventional reading of section 4.2 would be simply false: the probe is reading the locus, not the edit.

The three encoders are mechanistically consistent with one another. HyenaDNA’s response at ± 48 bp is about $6\times$ smaller than Carbon’s (rel Δ 0.0130 vs. 0.0759) and its AUROC is about chance. A weaker registered response carries a weaker signal, which is what one would expect and is worth stating because it makes the spread legible rather than mysterious. The BRCA2 transfer orders the same way, and is weak for both encoders that we measured on it: Spearman -0.265 for Carbon against -0.186 for NT-v2.

5.3 What this licenses, and what it does not

The claim we set out to make—*frozen genomic FMs register point edits informatively*—is false as stated, and table 3 is what stopped us from publishing it. The claim we can support is narrower and, we think, more useful:

Whether a frozen genomic FM’s pooled state usefully registers a point edit is an **encoder-specific** property. The predict-no-change trap, by contrast, is **universal**. And even for the best of the three encoders, the response still cannot be anticipated from the pooled state (section 6).

Two boundaries on this. First, we do not know *what* the spread tracks. All three encoders are at most 500M parameters, and objective, tokenizer, width, context length, and training data all vary together across them; a three-point comparison cannot attribute the 0.93-to-0.55 range to architecture rather than capacity, or to either rather than pretraining corpus. Second, and for the same reason, nothing here predicts how a larger or longer-context model would behave. Carbon is the existence proof that an informative interventional pooled edit response is *possible*; it is not evidence that it is typical. We take the honest reading to be that a group planning to build on this geometry must measure it on their own encoder first, because the measurement is cheap and the outcome is not predictable from architecture on the evidence we have.

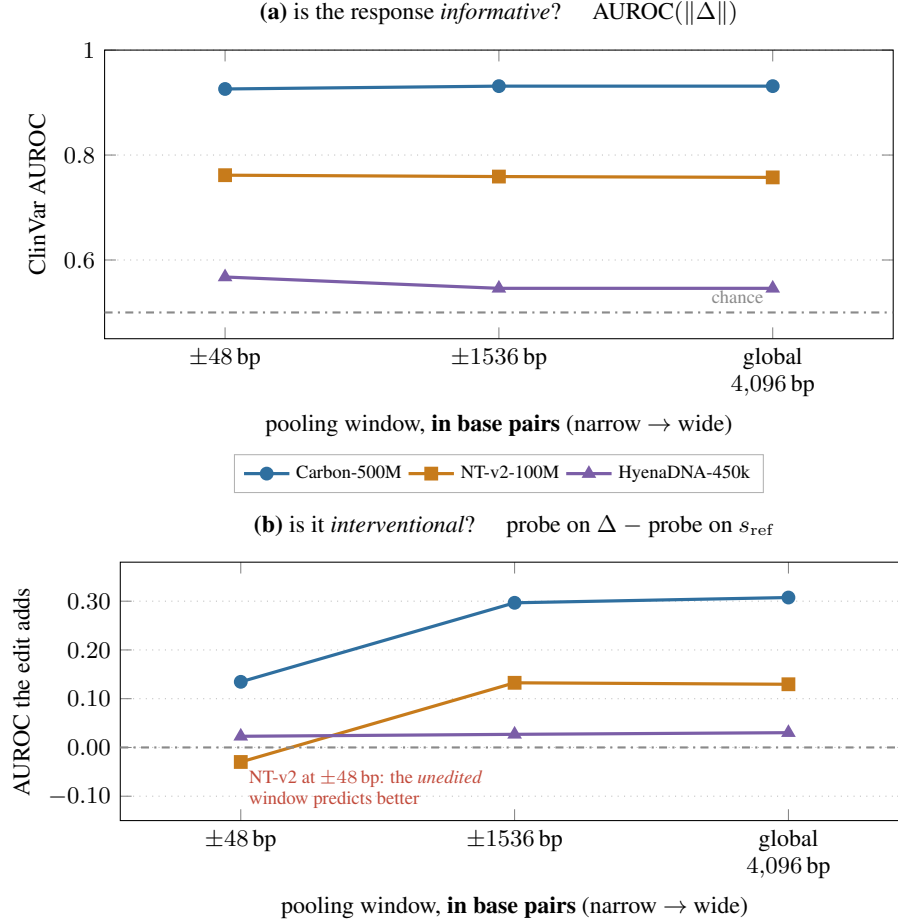


Figure 5: **The predict-no-change trap is a property of pooled genomic FM latents; the informative, interventional edit response is a property of *Carbon*.** The identical measurement, on the identical 12,993 SNVs, through three frozen encoders sharing neither training objective nor tokenizer (table 2, table 3). (a) AUROC of $\|\Delta\|$ for ClinVar pathogenic vs. benign. What is flat at 0.92–0.93 on Carbon is 0.76 on Nucleotide Transformer v2 and 0.55—near chance—on HyenaDNA. Informativeness does not replicate. (b) The fair interventional test: a supervised 5-fold probe on the displacement Δ minus the same probe on the reference state s_{ref} , which never observes the edit. Above zero, the edit tells you something the neighbourhood does not. Carbon clears it by +0.13 to +0.31 at every span; HyenaDNA by $\approx +0.02$, from a base of chance; and Nucleotide Transformer v2 goes *negative* at ± 48 bp (-0.0298), where its unedited window is the better predictor of the two. Not shown here, because it does not vary across the three: $\cos(s_{\text{ref}}, s_{\text{alt}}) \geq 0.993$ everywhere, reaching 1.00000 for HyenaDNA under global pooling (table 2). *Units*: the x axis is base pairs, not pool radius. Radius is counted in tokens, and Carbon and NT-v2 pack 6 bp per token against HyenaDNA’s 1, so HyenaDNA is run at radii 0/48/384/1536 to match the others’ 0/8/64/256 (section 3.4); comparing at equal radius would compare an ± 8 bp pool with a ± 48 bp one. The ± 384 bp span is omitted because the probe comparison is not reported at that span.

6 Predictability: The Core Negative Result

Carbon registers the edit (section 4), and of the three encoders we measured it is the only one whose registration is worth anticipating at all (section 5). A world model does not need the encoder to register the edit; it needs to *anticipate* that registration from the reference state and the action, because anticipating it is the only thing that saves the encoder pass. We now test that on Carbon—the best case available to us—under the protocol of section 3.5, at radius 8, the most favourable pooling width.

Table 5 and fig. 6 report the result. Reading the encoder’s true response gives AUROC 0.9259. The best learned predictor of that response, an MLP on (s_{ref}, a) trained with MSE, gives 0.6520. The substitution class alone, with no model at all, gives 0.6306. A 12-entry class-mean lookup table gives 0.6079. A predictor that ignores its inputs entirely gives 0.4939, i.e., chance, as it should.

The gap that matters is not the one between 0.6520 and 0.9259, though that gap is large. It is the one between 0.6520 and 0.6306. **The entire contribution of the learned model—all of it, including the pooled reference state, the nonlinearity, and the training—is the difference between 0.6520 and the 0.6306 you get for free by looking up which base changed into which.**

Conditioning on s_{ref} is worse than useless. The stronger statement is available from the linear model, and it is worth stating precisely because it is counterintuitive. Ridge on (s_{ref}, a) scores AUROC 0.5854—below the 0.6079 of the class-mean lookup, which ignores s_{ref} entirely and is a strictly less informed model. Direction agrees: $\cos(\hat{\Delta}, \Delta) = 0.1770$ for ridge against 0.2107 for the action-only lookup. Giving a linear predictor the pooled reference state does not merely fail to help; it degrades the prediction, because s_{ref} contributes 1024 dimensions of neighbourhood summary that are near-irrelevant to the response and dilute the one signal (the substitution class) that is relevant. The nonlinear cosine-trained MLP improves direction to 0.3228, so some structure is recoverable, but the MSE-trained model’s magnitude—the component that actually carries the label—sits at 0.1949 cosine and 0.6520 AUROC. That the harness is faithful is confirmed by an exact identity: ridge on the action alone reproduces the class-mean lookup to every digit ($\cos = 0.2107$, AUROC 0.6079), as it must, since one-hot ridge is class means.

The radius trend confirms the mechanism. If the cause is an information bottleneck in the pooled state, then predictability should degrade as the state is pooled more, while the ceiling stays flat. It does. At radius 8 (± 48 bp, the most local state) the MLP-MSE predictor reaches 0.6520 against a ceiling of 0.9259. At global pooling it collapses to 0.5400—essentially chance—with $\cos(\hat{\Delta}, \Delta) = 0.0575$, against a ceiling that is, if anything, *higher* at 0.9313. The ceiling is flat in the pooling width; the predictability is not. The wider the pool, the less the response can be anticipated, even though the response stays equally informative. That is precisely the signature of a bottleneck: the pooled state summarises the neighbourhood while discarding the local sequence context—which 6-mer token is being rewritten, and in what immediate surround—that determines the response.

The conclusion, stated plainly. The pathogenicity signal in the edit response is exactly the part that is not predictable from the pooled reference state plus the action. A latent world model over Carbon’s pooled states cannot recover it—not because the predictor is too small or the optimizer too weak, but because the conditioning information is not in s_{ref} . The efficiency thesis fails at its premise. And it fails in the most favourable case we could construct: Carbon is the one encoder of the three whose edit response is informative enough to be worth anticipating (section 5), so it is the one with the most to lose here. We tested predictability only on Carbon, and we do not extrapolate the negative to encoders we did not test—but an encoder whose response is uninformative offers a world model less to predict, not more. **There is no shortcut through the pooled latent: you must run the encoder on the edited sequence.**

7 Baselines and Scope

7.1 Quantitative function: where the measurement is weak

The 0.93 AUROC is on curated binary labels, and we do not want it read as a variant-effect result. Table 6 reports the harder test. On the BRCA2 saturation genome-editing assay of Sahu et al. [24] ($n=6,000$, radius 8), Spearman between $\|\Delta\|$ and the measured functional score is -0.265 (permutation $p < 0.001$). Transferring the direction learned on ClinVar gives $\rho = -0.155$.

These correlations are statistically real and practically weak, and the contrast with 0.93 is the honest content of this section. Two things differ between the evaluations. ClinVar is ascertainment-biased toward clear-cut variants—it is a curated record of what clinicians confidently called, so its positives and negatives are drawn from the tails. BRCA2 sGE, by contrast, spans all SNVs in one gene’s exons at fixed context, including the entire ambiguous middle, and asks for a graded quantity rather than a binary one. Separating curated extremes is a much easier problem than ordering a functional

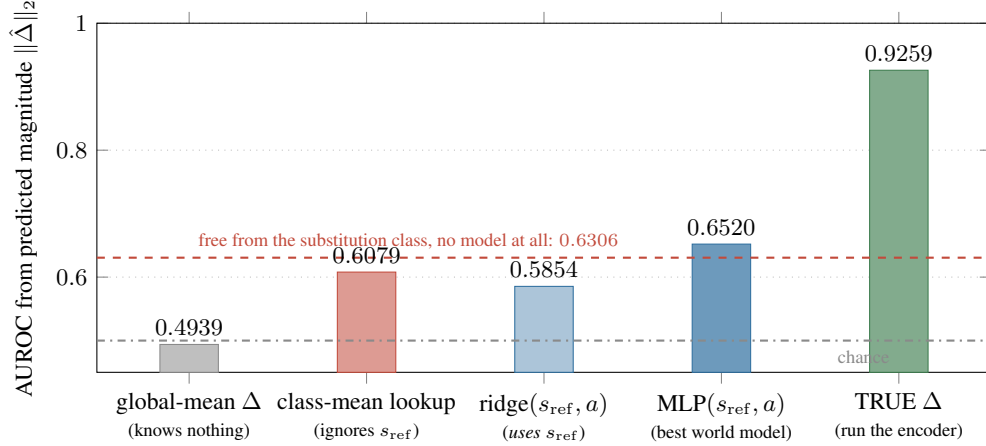


Figure 6: **The core negative result: the pathogenicity-bearing component of the edit response is exactly the component that cannot be anticipated.** All bars are AUROC for ClinVar pathogenic vs. benign, scored by the magnitude of a *predicted* displacement, at radius 8—the most favourable pooling width—under 5-fold cross-validation ($n=12,993$ SNVs). Reading the encoder’s true response gives 0.9259 (green). The best learned predictor of that response from the pooled reference state plus the action recovers 0.6520 (blue), barely above the 0.6306 available from the substitution class alone with no model whatsoever (red dashed line). Note the third bar: ridge, which *does* see s_{ref} , scores 0.5854—below the 0.6079 class-mean lookup, which does not. Conditioning on the pooled reference state is not merely uninformative here; under a linear model it is actively harmful. The information is registered by the encoder; it is not recoverable from the pooled state it leaves behind. Only the MSE-trained predictor’s magnitude is reported here: a cosine-trained predictor’s $\|\hat{\Delta}\|$ is scale-free and therefore meaningless as a magnitude (section 3.5).

Table 4: **The edit response is informative, and no trivial confound explains it.** ClinVar pathogenic-vs-benign AUROC from the displacement Δ , swept over pooling width, with every control we could construct from the same artifact. *All rows*: the raw cohort, which is contaminated by indels. *SNV-only*: restricted to the 12,993 single-nucleotide variants (899 pathogenic / 6,094 benign in ClinVar), excluding 1,007 non-SNV rows. *Within substitution class*: additionally scored only within each of the 12 substitution classes (covering $n=6,993$), removing the mutation spectrum. *Substitution only*: the spectrum with no geometry and no model. *Reference probe*: a linear probe on s_{ref} alone, which never observes the edit. *Chromosome held-out*: the same probe with chromosome-disjoint folds, testing for locus leakage. The pooling grid is ordered by width, not by the radius integer: $r=0$ denotes global-mean pooling over the whole 4,096 bp window.

Pooling	All rows	Edit-response Δ		Controls (no edit geometry)		
		SNV-only	Within sub. class	Substitution only	Ref. probe	Chrom. held-out
$r=8$ (± 48 bp)	0.9376	0.9259	0.9202	0.6306	0.6392	0.6354
$r=64$ (± 384 bp)	0.9362	0.9278	0.9220	0.6306	0.5741	0.5753
$r=256$ (± 1536 bp)	0.9323	0.9313	0.9245	0.6306	0.5539	0.5523
global (4,096 bp)	0.9301	0.9313	0.9240	0.6306	0.5500	0.5473

gradient. **This is not a competitive variant-effect-prediction method and must not be presented as one.** We report the BRCA2 number because it bounds what the geometry measurement is good for.

7.2 Relation to the model’s own likelihood, and the difficulty of our cohort

An AUROC of 0.92 from a frozen encoder’s displacement geometry is *higher* than the 0.6–0.8 typically reported for zero-shot Δ log-likelihood variant scoring with DNA language models [5, 19, 21]. A referee should not accept that at face value, and neither should we. There are two readings. Either our ClinVar slice is unusually easy, or embedding displacement genuinely beats likelihood

Table 5: **The edit response is not anticipatable from the pooled reference state.** Prediction of the displacement Δ from (s_{ref}, a) *without* encoding the edited window, at radius 8 (the most favourable pooling width), 5-fold cross-validation, $n=12,993$ SNVs. $\cos(\hat{\Delta}, \Delta)$ measures direction; $\text{AUROC}(\|\hat{\Delta}\|)$ measures whether the predicted magnitude carries the pathogenicity signal that the true magnitude carries. The cosine-trained MLP’s magnitude is omitted by construction, not by selection: a cosine objective is scale-invariant, so its $\|\hat{\Delta}\|$ is unconstrained and its AUROC would not be a fair test (section 3.5). Every learned predictor sits between the model-free baselines and the ceiling, far closer to the baselines; ridge, which sees s_{ref} , scores *below* the class-mean lookup, which does not.

Predictor of Δ from (s_{ref}, a)	$\cos(\hat{\Delta}, \Delta) \uparrow$	$\text{AUROC}(\ \hat{\Delta}\) \uparrow$
<i>Model-free references</i>		
Global-mean Δ (knows nothing)	0.0493	0.4939
Per-substitution-class-mean lookup	0.2107	0.6079
Substitution class alone, no model at all	–	0.6306
<i>Learned world models</i>		
Ridge(s_{ref}, a), λ swept $1-10^6$ (best 10^3)	0.1770	0.5854
MLP(s_{ref}, a), MSE loss	0.1949	0.6520
MLP(s_{ref}, a), cosine loss	0.3228	n/a (scale-free)
<i>Ceiling</i>		
TRUE Δ (run the encoder on the edited window)	1.0	0.9259

surprise on this task. The measurement that discriminates between them is Carbon’s own Δ log-likelihood scored on the *same* variants: it is the standard score behind those published numbers, so it doubles as a *difficulty calibration* of our cohort. We report it at radius 8, joined to the geometry on (chrom, pos, ref, alt), on $n=6,981$ ClinVar SNVs (893 pathogenic), in table 6.

Both readings turn out to be partly right, and we state each plainly because they are easy to mistake for a contradiction. They are not: the first is about our *cohort*, the second about our *quantity*.

(a) Our cohort is easy, and every AUROC in this paper is cohort-dependent. $-\Delta$ log-likelihood reaches AUROC 0.8945 on this slice. That is far above what the same score attains in the published zero-shot DNA-LM literature, and it is a fact about our data rather than our method. **Our ClinVar slice is easy, and that—not a superior score—is the reason our raw AUROC exceeds the literature’s.** ClinVar ascertainment is the obvious mechanism (section 9): the record contains what clinicians confidently called, so its positives and negatives are drawn from the tails. We therefore report 0.9255 as a cohort-dependent measurement and explicitly *not* as an absolute win over likelihood scoring; nothing here licenses a claim that this geometry is a better variant-effect predictor than Δ log-likelihood on any cohort but ours, and the weak BRCA2 transfer (section 7.1) points the same way.

(b) But on identical variants the geometry carries orthogonal signal. Calibration is not the whole story, because the two scores are computed on the same 6,981 variants and they do not agree with each other. $\|\Delta\|$ scores 0.9255 against the likelihood’s 0.8945. They are only weakly correlated: $\text{Spearman}(\|\Delta\|, \Delta \log \text{Lik}) = -0.2391$. Residualising $\|\Delta\|$ on $\Delta \log \text{Lik}$ in rank space—deleting everything the likelihood explains—still leaves AUROC 0.8529. And a rank ensemble of the two reaches 0.9482, above either alone, which is the signature of two partly independent measurements rather than one measured twice. The pattern survives the trinucleotide stratification of section 4.2: within context, $\|\Delta\|$ scores 0.9302 against the likelihood’s 0.9007 ($n=5,021$, 64 contexts). On BRCA2 the two separate further: Spearman with the functional score is -0.2649 for $\|\Delta\|$ against $+0.1124$ for $\Delta \log \text{Lik}$. Both are weak—this is the section that says so—but the likelihood carries almost nothing there.

The consequence is a scoping claim, not a performance claim. $\|\Delta\|$ **is not a restatement of likelihood surprise.** A frozen encoder’s pooled displacement and its next-token distribution register overlapping but distinct things about the same edit, which is why the embedding-distance paradigm [18] is not simply a noisy proxy for the $\Delta \log \text{Lik}$ paradigm [5, 19, 21], and why the geometry was worth dissecting rather than replacing with a likelihood. It is also why section 6 is ceiling-relative

Table 6: **Scope of the measurement beyond curated binary labels, and its relation to Carbon’s own likelihood.** The BRCA2 rows are the honest limit of the result: strong separation of curated pathogenic/benign extremes does *not* transfer to a quantitative functional score within a single gene. The lower block does two jobs at once (section 7.2). It *calibrates the difficulty of our cohort*: $\Delta \log$ -likelihood, the standard zero-shot DNA-LM variant score, reaches 0.8945 here against 0.6–0.8 in the published literature, so our ClinVar slice is easy and every AUROC in this paper must be read as cohort-dependent rather than as a method win. And it shows that, on identical variants, the geometry is nonetheless *not* a restatement of likelihood surprise: it scores higher, correlates with the likelihood only weakly, survives residualisation of the likelihood, and ensembles above either score alone. This is not a competitive variant-effect-prediction method and we do not present it as one (section 9).

Evaluation	Quantity	n	Value
ClinVar (curated binary)	AUROC($\ \Delta\ $), $r=8$, SNV-only	6,993	0.9259
BRCA2 sGE (quantitative)	Spearman($\ \Delta\ $, function), $r=8$	6,000	−0.265
BRCA2 sGE (quantitative)	Spearman, ClinVar-learned direction transfer	6,000	−0.155
<i>Relation to Carbon’s own likelihood; same variants, joined on (chrom, pos, ref, alt), $r=8$, fp32</i>			
ClinVar SNVs (893 path.)	AUROC($\ \Delta\ $)	6,981	0.9255
ClinVar SNVs (893 path.)	AUROC($-\Delta \log \text{Lik}$)	6,981	0.8945
ClinVar SNVs (893 path.)	AUROC($\ \Delta\ $ residualised on $\Delta \log \text{Lik}$)	6,981	0.8529
ClinVar SNVs (893 path.)	AUROC(rank ensemble of the two)	6,981	0.9482
ClinVar SNVs (893 path.)	Spearman($\ \Delta\ $, $\Delta \log \text{Lik}$)	6,981	−0.2391
within trinucleotide context	AUROC($\ \Delta\ $)	5,021	0.9302
within trinucleotide context	AUROC($-\Delta \log \text{Lik}$)	5,021	0.9007
BRCA2 sGE (quantitative)	Spearman($\ \Delta\ $, function)	6,000	−0.2649
BRCA2 sGE (quantitative)	Spearman($\Delta \log \text{Lik}$, function)	6,000	+0.1124

The within-context rows pool the 37 of the 64 trinucleotide contexts present that carry at least ten variants per class—the same 37 as table 1. The two differ only in cohort: table 1 covers 5,027 variants of the 6,993

ClinVar SNVs, while the rows here cover 5,021 of 6,981, twelve fewer because twelve duplicated (chrom, pos, ref, alt) keys collapse when the set is keyed for the likelihood join. The BRCA2 −0.2649 is the −0.265 above at four decimal places, not a second measurement.

and untouched by any of this: whatever $\|\Delta\|$ measures, and however easy the cohort, the part of it that carries the label is the part a world model cannot anticipate.

A numerical hazard we walked into, and that others will. Our first attempt at this measurement was invalid, and the failure mode is worth a paragraph because it is silent and the quantity is a standard one. Computed in bf16—the precision at which every other measurement in this paper runs—90.7% of the $\Delta \log$ -likelihoods came back *exactly* zero, and the survivors took only 18 distinct values, every one a multiple of 16. Nothing was wrong with the model, the data, or the code. A 4,096 bp window’s log-likelihood has magnitude $\approx 5,000$, and the spacing between adjacent representable bf16 values at that magnitude *is* 16: a single-base edit changes the window log-likelihood by far less than one unit in the last place, so the difference is annihilated before it can be formed. The reported run is fp32, on which the same pipeline yields 0 exact zeros, 9,115 distinct values, and a range of −486.0 to +36.3. **Do not accumulate or difference log-likelihoods in bf16.** We cite no number from the bf16 run anywhere in this paper. The general lesson is cheap to apply: before interpreting any distribution of differences, count its exact zeros and its distinct values—a large mass of exact zeros over a small set of evenly spaced levels is a float artifact, not a finding.

8 Why the Released Run Failed, and Why a Correct One Would Have Too

This project previously released v0.2.1-r1: an action-conditioned latent predictor over a frozen Carbon encoder, trained to predict $\hat{s}_{t+1}$ from (s_t, a) , benchmarked on ClinVar, BRCA2, and Trait-Gym, and published with content-addressed receipts binding every reported number to its bytes. Its results were withdrawn. This section preserves that record, and then makes the point that matters: the defects were real, but they are not the reason the idea could not work.

8.1 What was actually broken

A post-release audit established four defects, each verified against the released artifacts (table 8, table 7).

The normalization contract was ignored. The configuration set `normalize: true`, but the encoder returned raw pooled states while the predictor always returned a unit-norm vector. On the released rollout slices, mean source/target norms were 33.982/33.595 (phased) and 29.253/29.089 (synthetic), against a mean prediction norm of 1.000. Every L_2 loss, surprise score, and rollout distance therefore compared operands differing by a factor of ≈ 30 , and was dominated by scale rather than by content.

Pooling coordinates were mismatched. Training encoded source states with no edit locus supplied, which silently triggered global-mean fallback, while targets were pooled around the edit. Source and target states were thus in different coordinate frames. Separately, every “centered” pool omitted Carbon’s leading `<dna>` control token and was shifted one hidden token left, sometimes centering the pool on the control token itself.

The regularizer had no gradient. The run was labelled `phase2`, but the encoder remained frozen with no LoRA parameters, and its Gaussian KL term was computed only from frozen target states—constant with respect to every trainable parameter, contributing exactly zero gradient. No Phase-2 claim was tested by it.

The runtime was not self-contained. The pinned `Carbon tokenizer.py` performed an unpinned, network-capable transitive load of `Qwen/Qwen3-4B-Base`, so hashing the local tokenizer file did not define the tokenizer that actually ran.

Consequently the released L_2 losses, surprise scores, L_2 rollout metrics, and every conclusion drawn from them do not evaluate the intended method. The released cosine values (0.289 and 0.302 prediction-to-target, versus 0.998 and 0.991 source-to-target) remain reproducible historical measurements—cosine is scale-invariant—but they are confounded by mixed-scale, mixed-pooling training and by train/rollout distribution shift, and they establish neither capability nor mechanism. The run used seed 271828, 10,000 optimizer steps, and 80,000 samples; its ClinVar slices were $N=16$. We preserve these numbers in table 7 and table 8, and their identity in table 9, for auditability; we withdraw the interpretations.

8.2 The lesson the defects do not teach

That audit explains why *that run* was uninterpretable. It is silent on whether the idea was sound. It is tempting—and it was our first reading—to conclude that the engineering was at fault and that a corrected rerun would settle the question. Section 6 says otherwise.

Consider a hypothetical corrected `v0.2.1`: normalization applied consistently, source and target pooled at the same token-layout-derived centre, a gradient-bearing regularizer, a self-contained tokenizer, matched train and rollout distributions, and adequately powered slices. It would have failed anyway, for two compounding reasons that no amount of engineering removes.

First, its objective was degenerate. It trained $g(s_t, a) \rightarrow \hat{s}_{t+1}$ against a target with $\cos(s_t, s_{t+1}) = 0.99989$ at global pooling (section 4.1). The identity function attains that target to four decimal places. A corrected run would have converged to a near-copy of its input and reported a good-looking loss, and the loss would have measured nothing, because the discriminative content lives in a residual the metric barely weighs. Fixing the normalization bug would have made the numbers *coherent* without making them *informative*.

Second, and fatally, the conditioning information was absent. Even after moving to the correctly-posed displacement objective—which is what section 6 does, with a clean implementation on clean data—the best predictor of Δ from (s_{ref}, a) recovers 0.6520 of the 0.9259 available, barely above the 0.6306 that requires no model at all. The pooled state is an information bottleneck that discards precisely the local sequence context determining the edit response. This is a property of Carbon’s pooled frozen features—the very features that release was built on—and not of our optimizer, our

Table 7: **Historical variant-effect measurements from the withdrawn v0.2.1-r1 release; retained for auditability, invalid as a method comparison.** The GenoLeWM score was an L_2 residual between unit-norm predictions and raw target states whose norms were about 29–34 (table 8), so these rows do not evaluate the intended normalized-state method. The slices are additionally underpowered ($N=16$ per ClinVar slice, $N=32$ for BRCA2 and TraitGym). Δ is GenoLeWM – Carbon. Bold marks only the larger historical value.

Benchmark	Metric	GenoLeWM	Carbon 0-shot	Δ
ClinVar coding	Accuracy $\uparrow$	0.7500	0.6875	+0.0625
	AUROC $\uparrow$	0.7344	0.9219	−0.1875
	AP $\uparrow$	0.8530	0.9519	−0.0989
	Balanced acc. $\uparrow$	0.7500	0.6875	+0.0625
ClinVar non-coding	Accuracy $\uparrow$	0.4375	0.6875	−0.2500
	AUROC $\uparrow$	0.5625	0.8750	−0.3125
	AP $\uparrow$	0.6055	0.9144	−0.3090
	Balanced acc. $\uparrow$	0.4375	0.6875	−0.2500
BRCA2 saturation	Spearman ρ $\uparrow$	0.1492	0.4769	−0.3277
TraitGym Mendelian	Spearman ρ $\uparrow$	−0.0280	−0.0839	+0.0559

Table 8: **Historical rollout measurements and the state-norm audit that withdrew them (v0.2.1-r1).** The lower block is the direct evidence of the normalization defect: the predictor emitted unit-norm vectors while the encoder emitted raw states with norms near 29–34, so the L_2 columns compare incommensurable operands. The cosine columns are scale-invariant and remain numerically reproducible, but are confounded by mixed-scale training, mismatched pooling coordinates, and train/rollout distribution shift. Recall@ k was 1.0 for both model and baseline at $N=8$ and was non-discriminating. Section 8 argues that repairing all of this would not have rescued the run.

Setting	Cosine to s_{t+1} $\uparrow$ (confounded)			L_2 to s_{t+1} (invalid)		
	GenoLeWM	Source-state	Δ	GenoLeWM	Source-state	Δ
Phased haplotypes	0.2889	0.9978	−0.7090	33.32	2.13	+31.19
Synthetic edit chains	0.3016	0.9912	−0.6896	28.80	3.17	+25.64

Setting	Mean $\ s_t\ _2$	Mean $\ s_{t+1}\ _2$	Mean $\ \hat{s}_{t+1}\ _2$
Phased haplotypes	33.982	33.595	1.000
Synthetic edit chains	29.253	29.089	1.000

architecture, or our bugs. It would have been true of a bug-free implementation. It would have been true of a better-designed one.

This is why we consider the reframe honest rather than convenient. A retraction that says “we had bugs” generalises to nobody. The measurement says something stronger and more useful: this architecture—an action-conditioned latent world model over the pooled frozen states this release was built on—cannot deliver its premise, and the reason is information-theoretic rather than engineering. Two of its components generalise beyond Carbon and one does not, and we are careful about which: the degenerate cosine objective is a trap on every encoder we measured (section 5), whereas the claim that there was a rich response to anticipate in the first place is one only Carbon earns. That converts a local embarrassment into a result other groups can act on, and it is the reading the data supports.

9 Limitations

We state these as constraints on the claim, not as caveats to it. Several are severe.

The claim is about pooled states only. We show that the edit response cannot be anticipated from a pooled reference state plus an action. Pooled states are what latent world models actually use, which is why the result bites. But we have *not* shown that a token-level world model is impossible,

and nothing here argues that it is. A predictor operating on the full hidden-state sequence retains the local context that pooling discards, and is the natural follow-up. Our result draws a boundary; it does not close the area behind it.

Three encoders, none larger than 500M. An earlier draft of this paper carried a limitation headed “one encoder”, conceding that a single-model result was an anecdote about Carbon until replicated. Section 5 replaced that concession with a measurement, and the measurement changed the claim rather than confirming it: the predict-no-change trap replicated on all three encoders, and the informative, interventional edit response did not replicate on either of the other two. What was a caveat is now a result, and the paper is scoped accordingly—Carbon by name wherever the response is called interventional.

What remains untested is the direction the replication did *not* go. All three encoders are at most 500M parameters and share our 4,096 bp window, so we cannot say whether a larger or longer-context model behaves like Carbon or like HyenaDNA. Nor can a three-point comparison attribute the 0.9313-to-0.5460 spread to any single cause: objective, tokenizer, width, context length, and pretraining corpus all covary across these models, so the spread may track capacity or training objective rather than architecture as such. **We do not claim to know which**, and a reader who wants an answer should measure their own encoder rather than interpolate ours. Carbon is additionally held at one revision, one layer, and one window length; that much of the original limitation stands.

The action space is 12 substitution classes. Because every variant sits at the window centre by construction, position carries no information and the action reduces to which base changed into which. A richer action space—position, local sequence context, the identity of the rewritten 6-mer—is not tested here. This cuts against us in a specific and important way: a predictor given the local context we withheld might do better than 0.6520. We regard this as the most credible route to overturning our negative result, and it is the same route as the token-level follow-up above.

The 6-mer token pair is not fully controlled. Our trinucleotide control (table 1) conditions on ± 1 bp, but Carbon’s tokenizer packs 6 bp into a token, so the exact ref-6-mer $\rightarrow$ alt-6-mer token-pair confound survives it. The natural fix, a ± 2 bp (5-mer) stratification, is too sparse to be powered at 899 pathogenic variants, and we do not report an underpowered version of it. The trinucleotide result therefore bounds the k -mer explanation without eliminating it.

ClinVar ascertainment bias, and it is measured, not merely suspected. ClinVar records what was confidently classified, which biases it toward clear-cut variants at the tails of the effect distribution. We can put a number on how much this flatters us: Carbon’s own Δ log-likelihood, the standard zero-shot DNA-LM variant score, reaches AUROC 0.8945 on our slice against the 0.6–0.8 it attains in the published literature (section 7.2). **Our slice is easy**, and that—not the merit of the geometry—is why our raw AUROCs exceed the literature’s. AUROC 0.9259 should be read as separation of curated extremes on a favourable cohort, not as accuracy on the variants that are clinically difficult, which are exactly the ones anybody needs help with. Every AUROC in this paper is cohort-dependent and none of them are comparable to a benchmark number.

BRCA2 is one gene. The quantitative-function evaluation ($\rho = -0.265$) covers a single gene’s exons in a fixed context, from a single assay. It bounds the claim in the direction of weakness, and it should not be extrapolated in either direction to other genes or assays.

Not a variant-effect method. We report AUROC against controls to characterize a representation, not to compete on a benchmark. We do not compare against tuned VEP methods, we do not tune ours, and the BRCA2 result indicates it would not compete.

No clinical claim of any kind. Nothing in this paper is a clinical, diagnostic, or decision-support tool or claim. No pathogenicity label enters any encoder, and no output here should inform any decision about any person.

10 Conclusion

We measured the edit-response geometry of frozen genomic foundation models and found a clean dissociation. Carbon-500M *registers* a single-base edit richly: the pooled displacement separates ClinVar pathogenic from benign SNVs at AUROC 0.9259, and it does so interventionally—not via the mutation spectrum (0.6306 with no geometry), not via the exact local sequence context (conditioning on the full trinucleotide context costs 0.0059), not via the genomic neighbourhood (0.6392 from a probe that never sees the edit, with no locus leakage), and not via a local token swap (distal and local response directions agree at only $\cos = 0.4714$). Carbon propagates the edit through attention, and the signal survives at every pooling width, even where the edit is cosine-invisible (0.99989). It is not a restatement of the model’s own likelihood either: on identical variants it beats $\Delta \log$ -likelihood (0.9255 vs. 0.8945), correlates with it at only $\rho = -0.2391$, and survives residualising it out (0.8529). The AUROCs are cohort-dependent and the cohort is easy—the likelihood’s 0.8945 says so—so none of this is a variant-effect claim.

Two of those sentences have Carbon as their subject on purpose. Run through two encoders sharing neither objective nor tokenizer, the same measurement on the same variants gives 0.7574 and 0.5460, the latter near chance, and Nucleotide Transformer v2’s edit response is beaten at ± 48 bp by a probe on its own unedited window. Whether a frozen genomic FM’s pooled state usefully registers a point edit is an *encoder-specific* property. The predict-no-change trap is not: $\cos(s_{\text{ref}}, s_{\text{alt}}) \geq 0.993$ on all three encoders at every pooling width, reaching 1.00000. The trap is what generalises; the signal is what does not.

And on Carbon, where the response is worth anticipating, it cannot be *anticipated*. The best learned predictor of the response from the pooled reference state plus the action recovers AUROC 0.6520 of the 0.9259 available, against 0.6306 free from the substitution class alone. The reference state buys almost nothing, and a linear model conditioned on it does worse (0.5854) than a lookup table that ignores it (0.6079); predictability degrades toward chance as pooling widens while the ceiling stays flat. The pathogenicity-bearing component of the edit response is exactly the unpredictable one.

The consequence for the architecture we set out to build is unambiguous, and negative. An action-conditioned latent world model over Carbon’s pooled features cannot deliver its premise, because the premise is that you can skip the encoder pass on the edited sequence, and the encoder pass is where the signal is created—and Carbon is the most favourable of the three encoders we measured. Our own withdrawn v0.2.1-r1 run had genuine implementation defects, but the defects were not the reason it failed; a correct implementation would have failed too, and now we can say why. The information exists. It is not where a latent world model needs it to be. **There is no shortcut through the pooled latent.**

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A Notation

Symbol	Meaning
$w_{\text{ref}}, w_{\text{alt}}$	reference / edited DNA window (4,096 bp, variant at centre)
$\text{enc}(\cdot)$	Carbon-500M (rev. 5d31d59b), layer 20, mean pool at radius r , raw
$s_{\text{ref}}, s_{\text{alt}}$	pooled reference / edited state, $\mathbb{R}^{1024}$
$\Delta = s_{\text{alt}} - s_{\text{ref}}$	the edit response (displacement); the object of study
$\text{rel}\Delta$	$\ \Delta\ _2 / \ s_{\text{ref}}\ _2$, relative displacement
r	pool radius in tokens ; 6 bp per Carbon token, so $r=8, 64, 256 \equiv \pm 48, \pm 384, \pm 1536$ bp. HyenaDNA is 1 bp per token and is run at $r=48, 384, 1536$ to match in bp (section 3.4). All cross-encoder comparisons are reported in base pairs, never in r .
$r=0$	global-mean pooling over the whole window (the <i>widest</i> setting)
a	the action: one of 12 substitution classes
$g(s_{\text{ref}}, a) \rightarrow \hat{\Delta}$	the predictor under test; the “shortcut”
n	14,000 variants; 12,993 after SNV restriction; 6,993 ClinVar SNVs

B Artifact Identity and Provenance

Table 9: **Artifact identity and provenance.** The upper block identifies the measurement reported in this paper; the lower block identifies the withdrawn release discussed in section 8, retained so that its historical numbers stay bound to the bytes that produced them. The two lineages are disjoint: no result in section 4 or section 6 depends on any v0.2.1-r1 artifact.

Field	Value
<i>Edit-response measurement (this paper)</i>	
Encoder	HuggingFaceBio/Carbon-500M, rev. 5d31d59b, bf16, frozen
Representation	layer 20; mean pool; window 4,096 bp
R1 job	6a565de7b1669a49bf07274b (A100-large)
R1 code commit	e23fdf9
Analysis commit	82eb3c2 (tools/research/edit_response_analysis.py)
Predictability commit	187a1c5 (tools/research/delta_predictability.py)
Cohort	14,000 GRCh38-grounded variants $\times$ 4 pool radii = 56,000 rows; 0 skips
FASTA ref-base check	12,993/12,993 SNVs matched
<i>R5: Carbon Δ log-likelihood (section 7.2)</i>	
R5 job	6a5744c2b1669a49bf073b77, run r5-delta-loglik-fp32-edd9b5a
Precision	fp32. The earlier bf16 run r5-delta-loglik-82eb3c2 is invalid (90.7% exact zeros over 18 distinct values) and no number from it appears in this paper.
fp32 sanity	0 exact zeros; 9,115 distinct values; range -486.0 to $+36.3$
<i>R6: two further encoders (section 5); analysis code run unchanged</i>	
NT-v2 job	6a57474485d9643ce16d3a4a; data r6-nt-v2-693571d
HyenaDNA job	6a57474485d9643ce16d3a4c; data r6-hyenadna-693571d
R6 cohort (each)	the same 12,993 SNVs $\times$ 4 pool radii = 51,972 rows; 0 skips
Pooling	bp-matched: HyenaDNA at radii 0/48/384/1536 (1 bp per token) against Carbon’s and NT-v2’s 0/8/64/256 (6 bp per token)
Data	https://huggingface.co/datasets/abdelstark/geno-lewm-edit-response
Data files	r1-edit-response-e23fdf9/edit_response_embeddings.parquet, r1-edit-response-e23fdf9/edit_response_summary.json, r1-edit-response-e23fdf9/variants.jsonl
Code	tools/research/edit_response_spectroscopy.py, tools/research/edit_response_analysis.py, tools/research/delta_predictability.py, tools/jobs/edit_response_run.sh, tools/jobs/delta_loglik_run.sh
<i>Withdrawn release (section 8; historical only)</i>	
Model release	geno-lewm-v0.2.1-r1
Model id	sha256:cddb8f3b9671...f5f3ed6af4
Dataset snapshot	geno-lewm-data-v0.2.1-r1
Commit	d9b06815cf8e...5aa4154d
Hardware	NVIDIA H200 (Linux x86_64, glibc 2.35)
Suite outputs	28 / 28 verified at the byte level (semantics unverified)

C Citation Verification Notes

Bibliography metadata was checked against publisher pages, DOIs, or arXiv where available. The following caveats apply and should be resolved before any camera-ready version.

- **Carbon has no archival publication.** There is a Hugging Face model card and a tech-report.pdf in github.com/huggingface/carbon; no arXiv or journal version exists. We cite it as a technical report and pin model revision 5d31d59b. A moving GitHub artifact is a legitimate reviewer objection; the tech report should be archived.
- **LeJEPA attribution and mechanism.** LeJEPA is by Balestrieri and LeCun [4], and its estimator is SIGReg (random one-dimensional projections with characteristic-function matching). Our withdrawn release used a closed-form Gaussian KL inspired by the isotropic-Gaussian rationale; it is *not* SIGReg, and section 2 states this.

- **BRCA2 source disambiguation.** There are two distinct Nature-2025 BRCA2 saturation genome-editing papers. We cite Sahu et al. [24], DOI 10.1038/s41586-024-08349-1, and *not* the separate study at DOI 10.1038/s41586-024-08388-8.
- **MaveDB accession.** The score set `urn:mavedb:00001242-a-1` is the identifier recorded by our data pipeline for the BRCA2 sGE scores. Its mapping to Sahu et al. [24] should be confirmed directly on mavedb.org before assertion.
- **Author lists.** Larey et al. [17] is confirmed as arXiv:2602.17162 (February 2026) and its author list was retrieved from the arXiv API. Assran et al. [2], Oquab et al. [22], and Rubin et al. [23] have truncated author lists.
- **Unconfirmed venue.** Zhou et al. [27] is confirmed as arXiv:2411.04983; its ICLR-2025 acceptance is inferred from OpenReview and not confirmed.
- **Unconfirmed DOIs.** The Brixi et al. [7] and Avsec et al. [3] entries cite bioRxiv DOIs, which are solid; their journal DOIs were not confirmed and are not cited.